Object Oriented Programming - Intro

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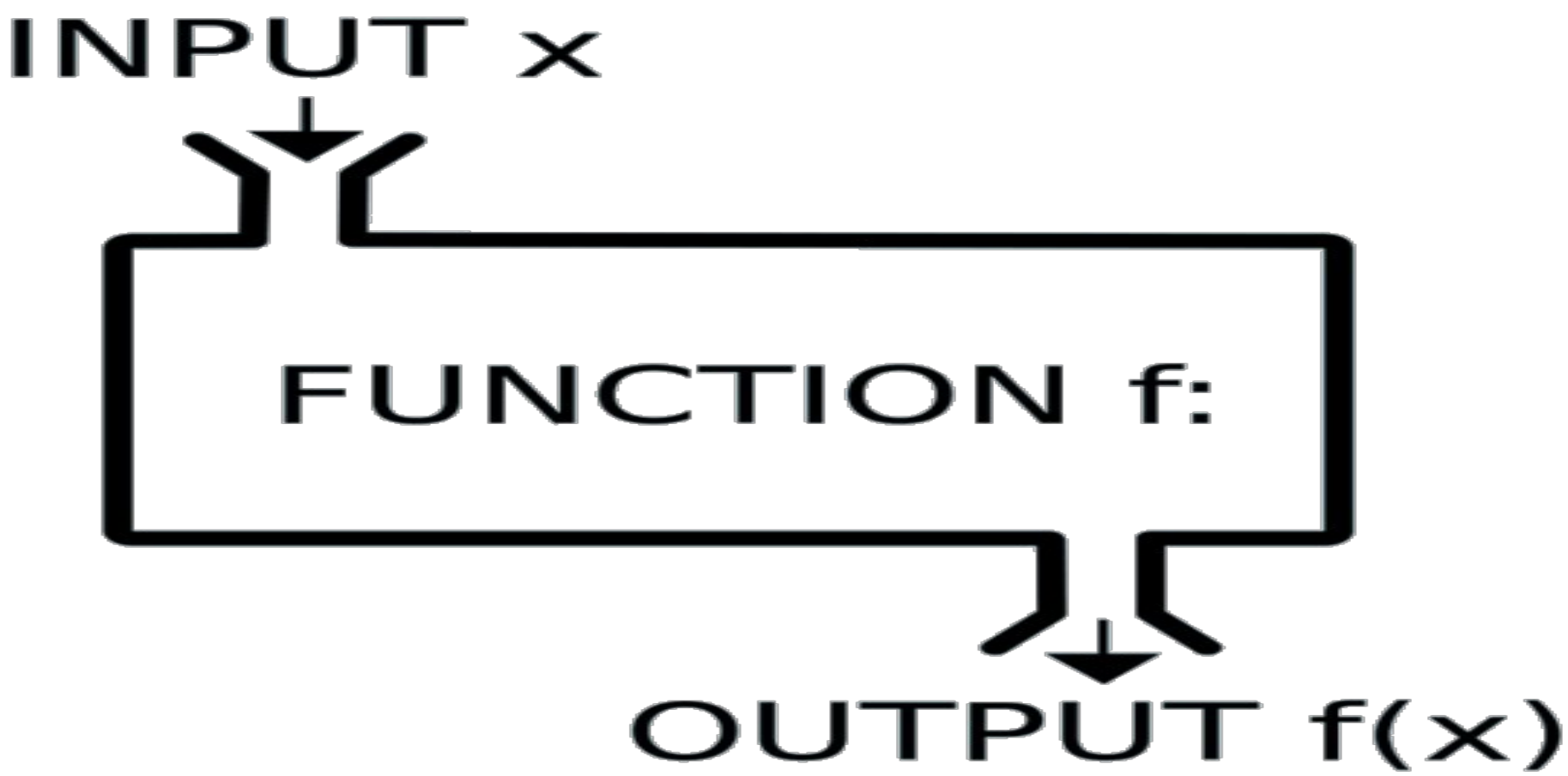
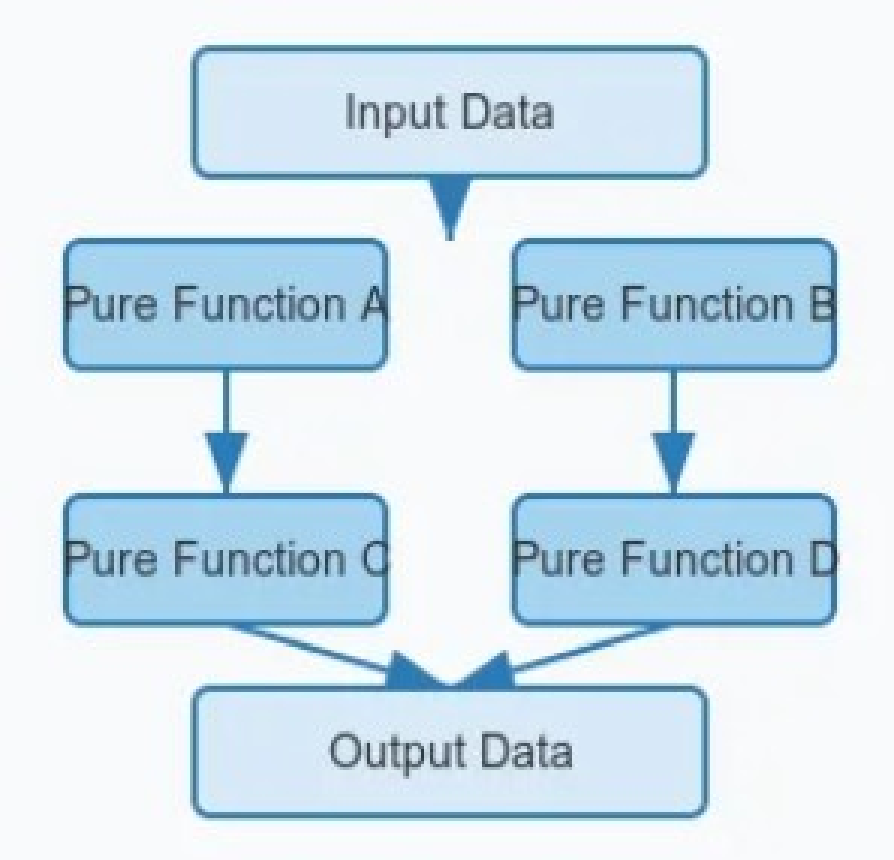
Functional Programming :: what we HAD ???



Functional Programming :: About




```
// global data
int balance;

void Calculate(int amount) {
    balance += amount;
}

void withdraw(int amount) {
    if (balance >= amount) {
        balance -= amount;
    }
}
```

Functional Programming :: Features

Based on Functions/Procedures, many if/else and recursion

Use shared GLOBAL data

Functions used/update shared data

*** Good for simple applications

*** Faster Execution

*** Best for :

- Operating Systems
- Embedded Systems
- Utility Applications

Languages:

- C, Pascal, Fortran, Early Basic

Functional Programming :: Issues

Problem to solve become complex

Global State CHAOS * WHO changed the Global value “BALANCE”**

One Function can break the other by doing wrong update to Global Data

Difficulty to monitor/track-down who own the data and who modified it

Tight Couple – *All Functions depends on Shared Data. No Clear control/ownership.*

Object Oriented Programming :: Intro

When the systems to develop is complex & BIG, issues arises such as

1. code complexity
2. difficulty of maintenance
3. difficulty of extend.

Main areas focused on

Who owns data and who is allowed to change it...

Data Protection

- data and data control code stay together

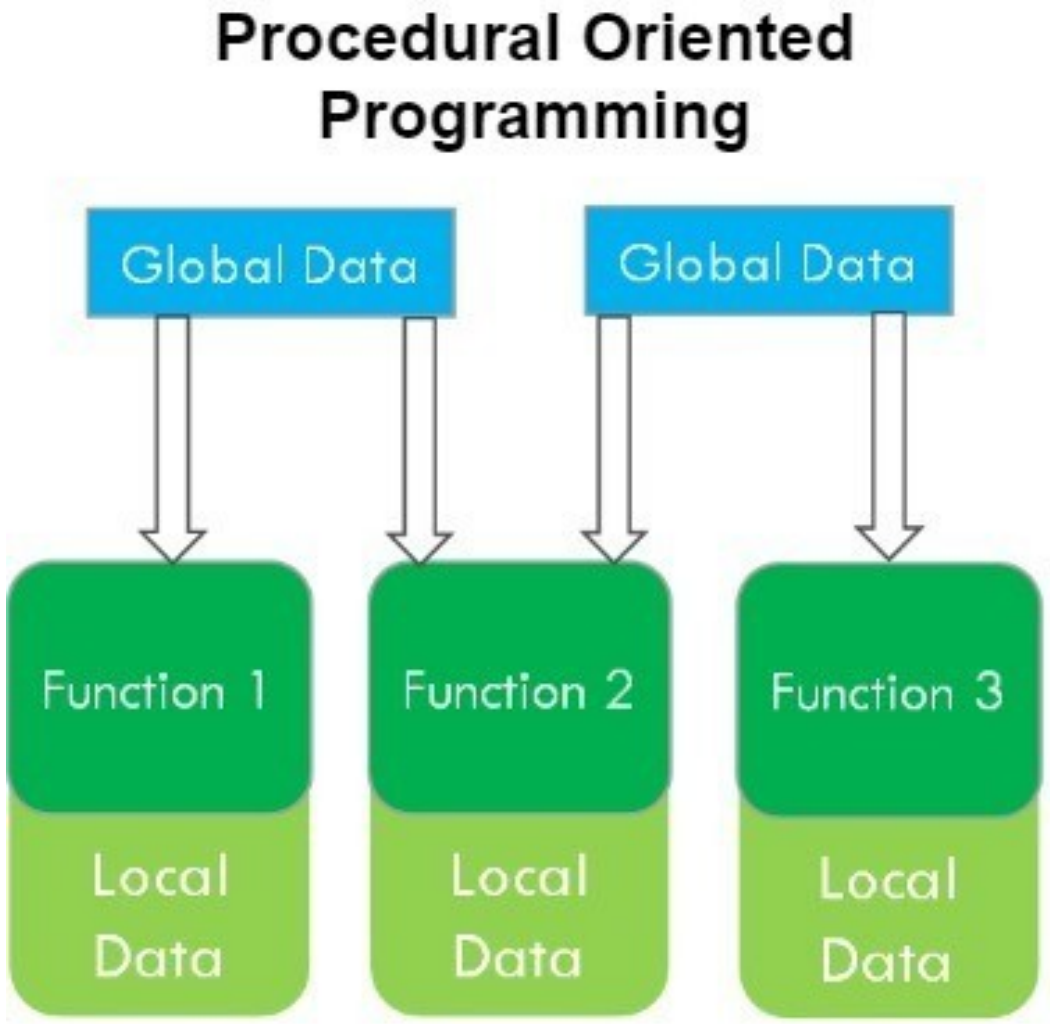
eg:- account class knows how to withdraw money and limit checks.

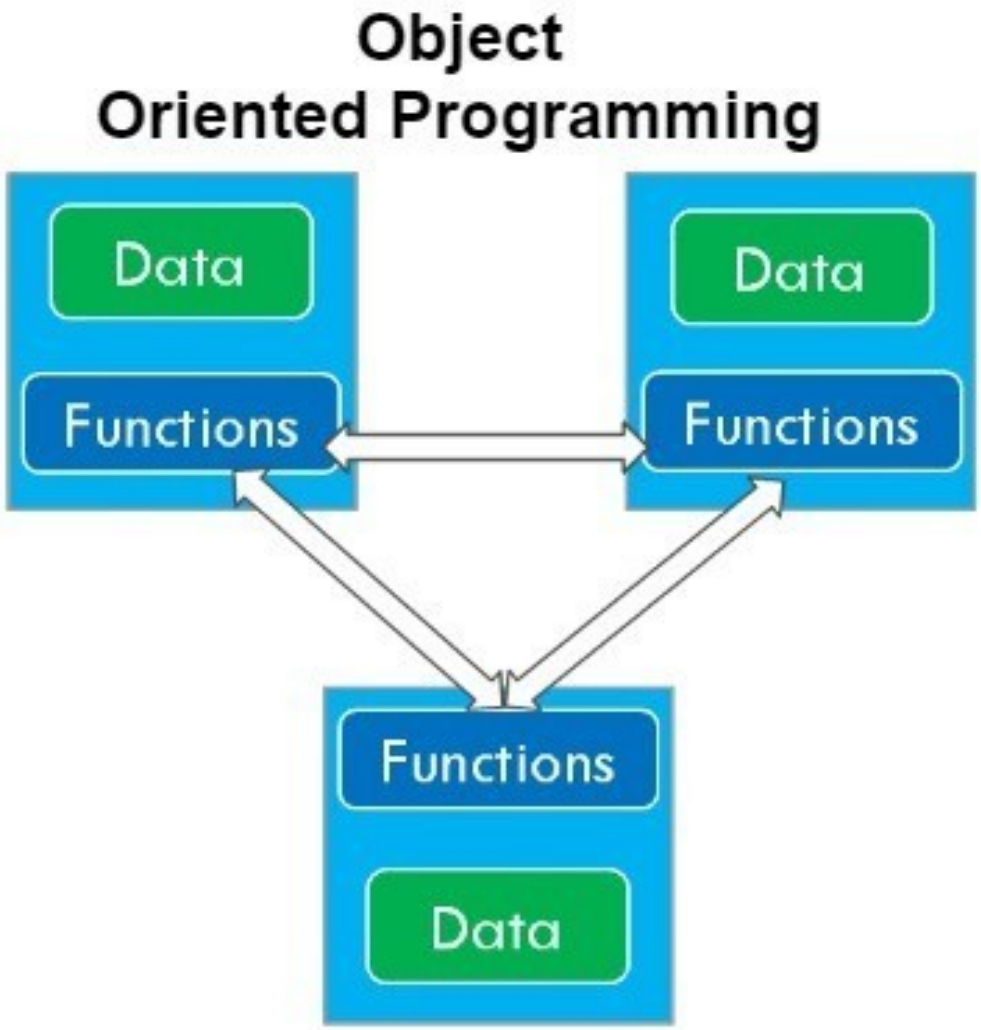
Reduce parameter passing

- account.withdraw(amount):
- account.deposit(amount):

Decoupling entities

Object Oriented Programming :: What to be done





Object Oriented Programming :: Idea

Idea of"Object" in programming emerged in late 1960 and early 1960s by AI group of MIT, US.

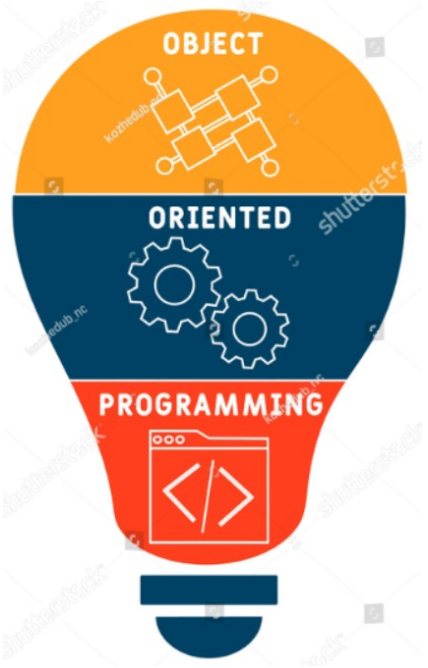
Sketchpad → 1st application with OOP in 1963 by Ivan Sutherland at MIT

That is for his PhD. Won several awards for it.

Ivan Sutherland:

AKA Father of Computer Graphics

Defined "Object" and "Instance" terms in his report.



In 1961-1967, Simula (programming language) was introduced in Norway

It has "classes", "inheritance" and "dynamic binding"

And it goes on …..

In 1980 - > C++, Objective-C, Eiffel…

In 1990 - > Visual Fox Pro, Delphi….

Object Oriented Programming :: 4 Pillars

Based on main 4 pillars

1. *Encapsulation (safety)*

   ”Protect the data. No direct access.”

2. *Abstraction (simplicity)*

   ”Show only what is required”

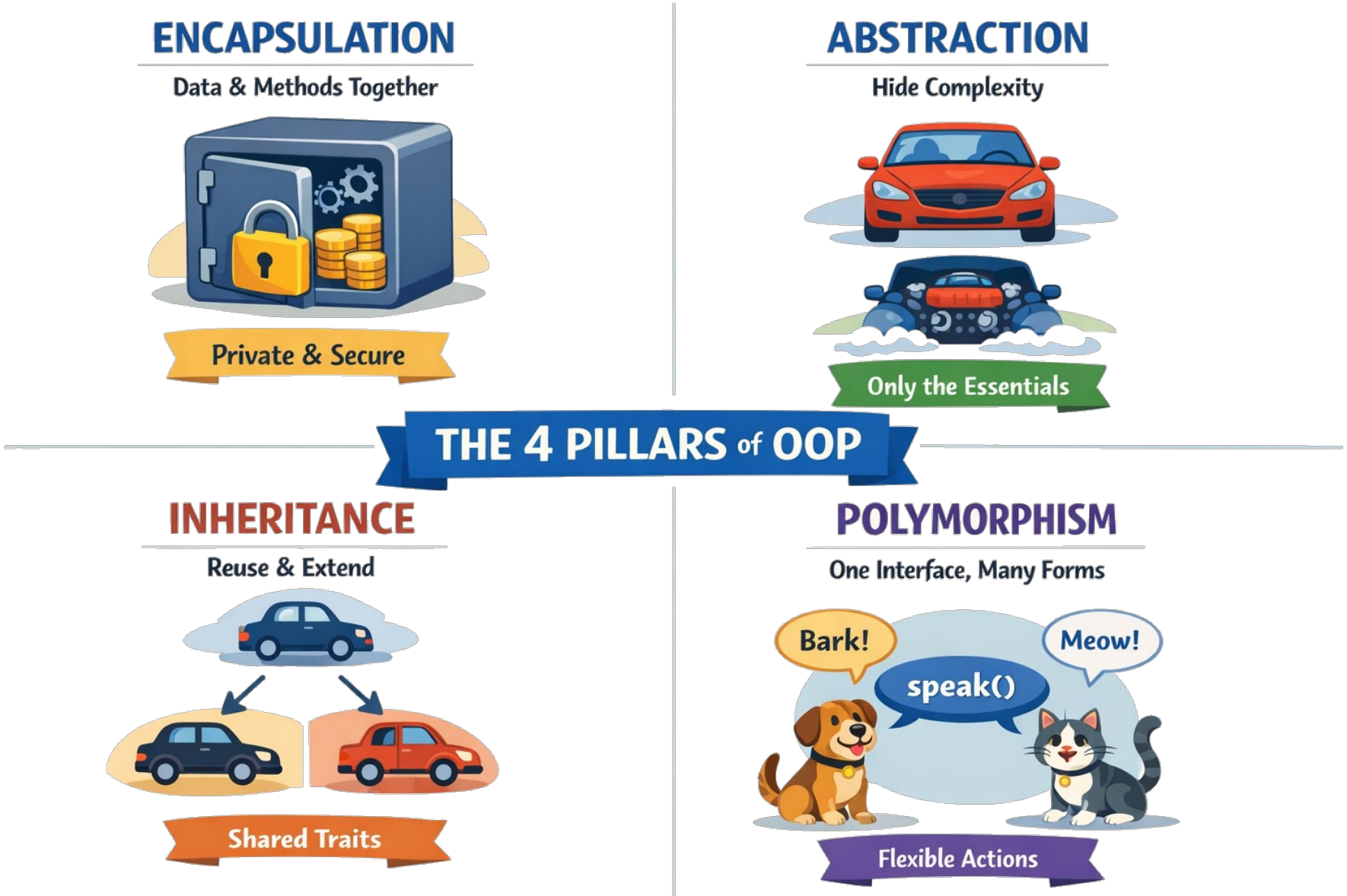
3. *Inheritance (efficiency)*

   ”Reuse what is already exists”

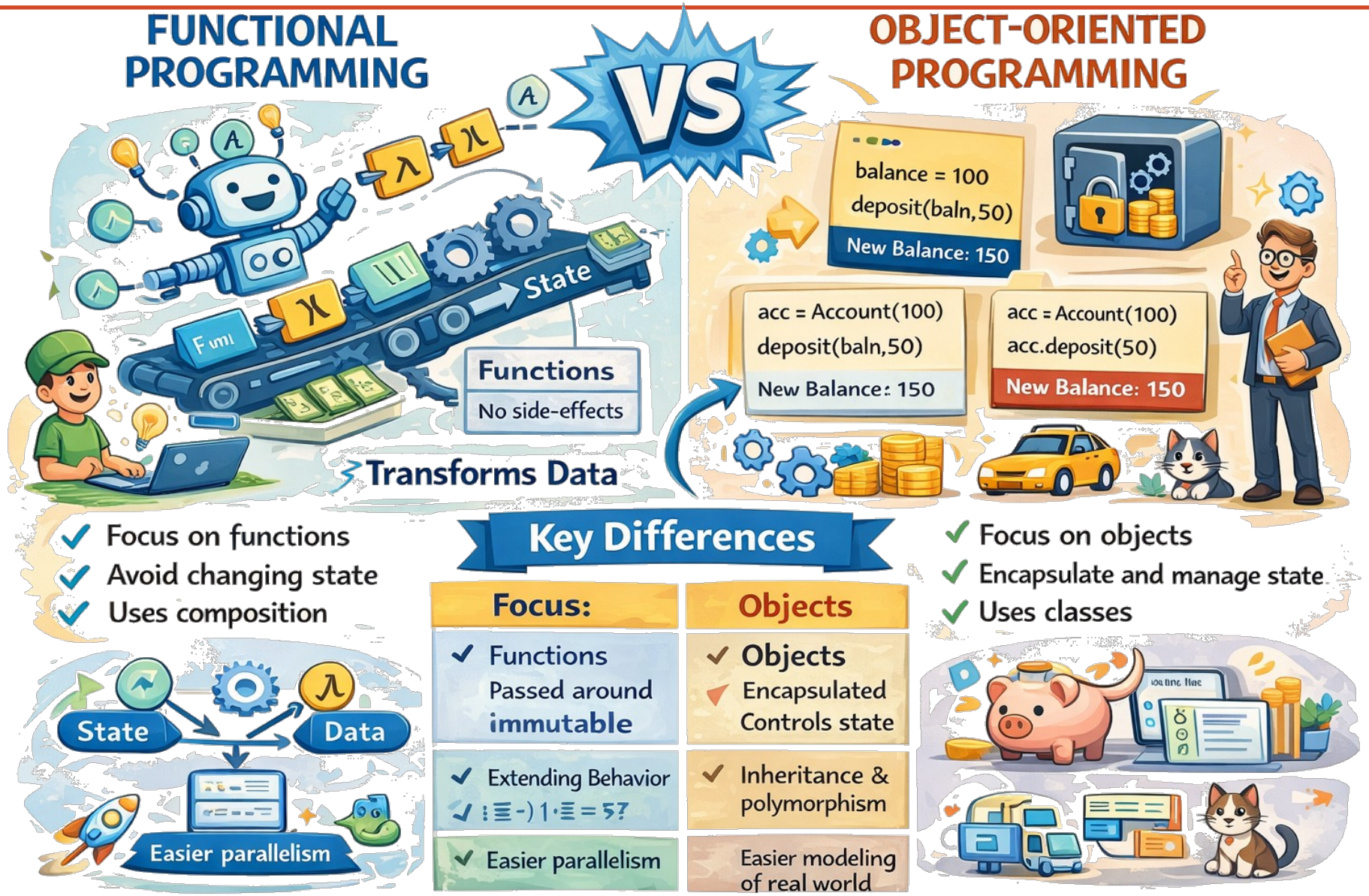
4. *Polymorphism (flexibility)*

   ”One action, many forms”

Object Oriented Programming :: 4 Pillars



Object Oriented vs Functional Programming



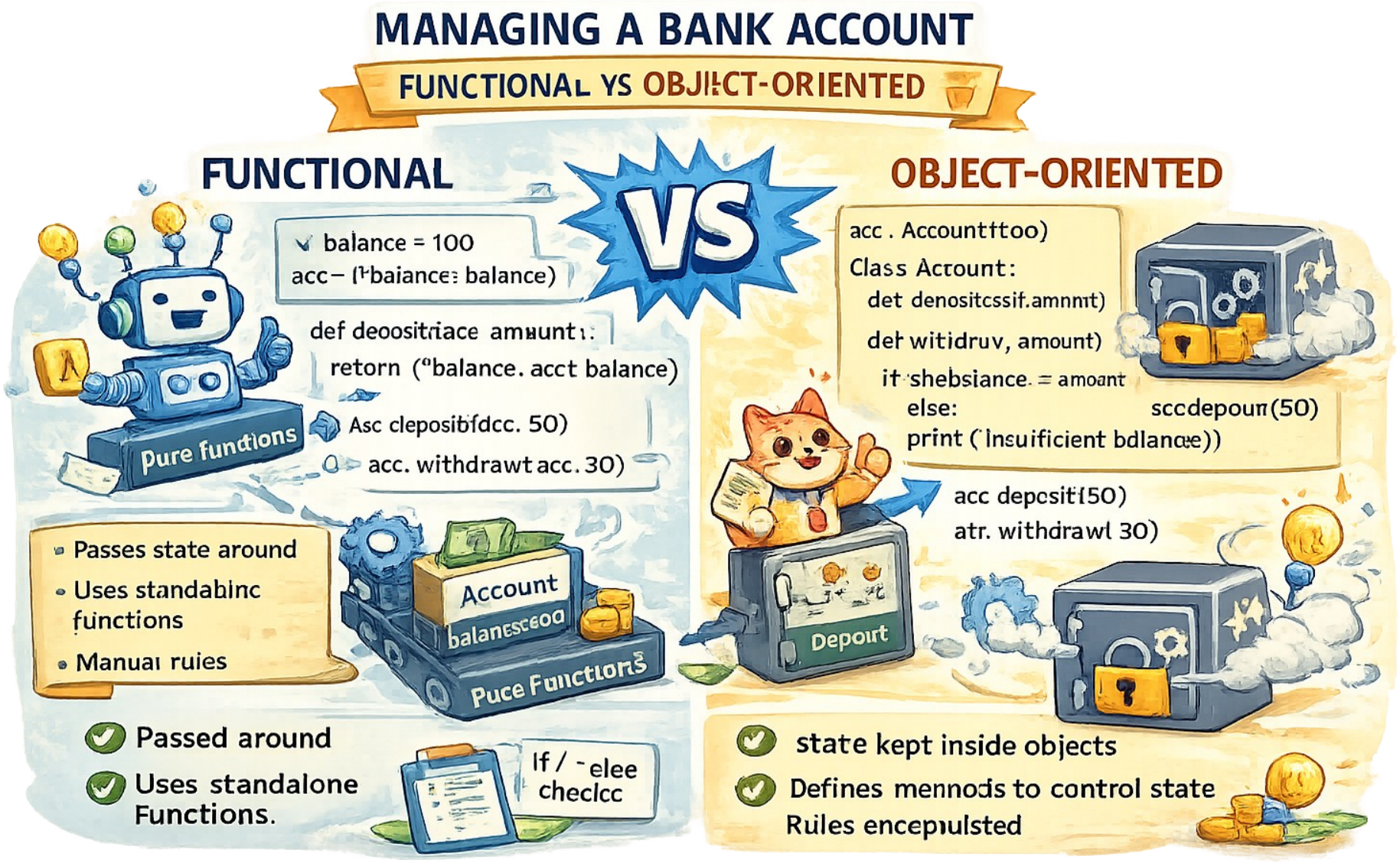
Object Oriented vs Functional Programming

FP is about “what to compute” — functions and transformations.

OOP is about “who owns the data and behavior” — objects represent real-world entities.

FP shines in: calculations, transformations, parallel processing, stateless systems.

OOP shines in: large systems, managing state, real-world modeling, team projects.

*** Modern software often uses both: FP inside methods of objects (hybrid approach).

Object Oriented vs Functional Programming - Sample